

Adaptive Modulation for Long-Range Underwater Acoustic Communication

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Abstract—Long-range underwater acoustic communication (LR-UWAC) refers to the peer-to-peer transmission of messages for distances of tens to hundreds of km. It is a key enabling technique for applications such as control over unmanned underwater vehicles for long-term surveying. While underwater acoustic communication over shorter ranges is an established technique, this is not the case for LR-UWAC. This gap is mostly due to channel uncertainties: in the absence of feedback from the receiver and due to the long transmission range, channel state information (CSI) at the transmitter may not reflect the actual channel. In this paper, we propose an adaptive approach to pre-set the modulation scheme for LR-UWAC. This is a channel classification approach which, based on environmental information and on prior training on various channel types, predicts the best modulation scheme for the expected channel. Our classification procedure is trained to identify the channel's important features. Thus, compared to a direct decision approach, it becomes less sensitive to possible mismatches of environmental information. Our numerical simulation and sea experiment show that our approach successfully identifies the best modulation scheme based on the environmental information - even when the information is biased or only partially available.

Index Terms—long-range underwater acoustic communications, channel simulation, channel classification, adaptive modulation.

I. INTRODUCTION

LONG-RANGE underwater acoustic communication refers to acoustic communication over distances of tens to hundreds of km. It is an essential technique for wide-range and long-term tasks carried out by manned or unmanned vehicles such as marine environmental monitoring, geological surveys, data collection of biological resources [1], [2], or even for acoustic tomography to measure sound speed and water temperature over long ranges [3]. Furthermore, LR-UWAC

can also be used for under-the-ice operations, geo-location, and communications to submarines [4]–[6].

While sound at low frequencies propagates well over long ranges, LR-UWAC is considered challenging due to its complex channel impulse response. Specifically, the channel's delay spread can be on the order of seconds, the multiple arrivals from boundary reflections are often non-separable, and the propagation loss is highly complex and simplified assumptions of, e.g., a ray-tracing model cannot be made [7]. An example of a channel impulse response estimated in our sea experiment (see details in Section III) is shown in Fig. 1. We observe a delay spread of more than 300 ms, and multiple non-separable arrivals of roughly the same power attenuation level. Such a long channel requires the implication of a very long equalizer. Moreover, the particular channel shown in Fig. 1 does not show evidence of a sparse reflection arrival pattern; thus, techniques such as compressive sensing are not suitable.

It is feasible to predict the channel with acoustic propagation modelings, through, e.g., the parabolic equation (PE) model [8] or the normal modes model [9]. However, these models are sensitive to environmental information. Still, in terms of communication, these models provide some information about the expected signal-to-noise-ratio (SNR) and the channel impulse response. Therefore, such information is useful to determine the system's preferred setting, i.e., the locations and depth of the transmitter and receiver, to evaluate the communication system's availability, and to pre-set the transmitted signal's modulation type [10]. Yet, with no feedback from the receiver, the accuracy of this latter evaluation highly depends on the quality of the environmental information. Improving the decision robustness based on such possibly mismatched information is the focus of this paper.

Despite the challenges of LR-UWAC, good performance was obtained in sea experiments in previous works. Freitag analyzed the transmitted m-sequence signal in the Acoustic Thermometry of Ocean Climate (ATOC) project as binary phase-shift keying (BPSK) communication signal, thereby demonstrating the feasibility of basin-scale LR-UWAC [11] for a range of up to 3,250 km. Later, more advanced modulation schemes, such as quadrature phase-shift keying (QPSK) [12], orthogonal frequency-division multiplexing (OFDM) [13], and 16 quadrature amplitude modulation (16-QAM) [14] were used in LR-UWAC. Communication ranges from 550 km to 700 km were achieved, and the data rate could go up to 400 bps. For all implementations, arrays of multi-hydrophones were used to compensate for the high propagation loss. The

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experiments were mostly performed in deep water, whereas LR-UWAC in shallow water is seldom reported.

In this paper, we do not propose another modulation type or receiver structure for LR-UWAC. Instead, we focus on the challenge of how to determine the modulation type with the limited available environmental information, hence the low accuracy of the predicted channel impulse response. The scope of this work is motivated by the understanding that, for LR-UWAC, the number of channel types or channel impulse response structures are limited. We draw our method from the concept of adaptive modulation, which adaptively changes the transmission strategies according to the actual channel conditions, thereby achieving improved communication performance [15], [16]. However, adaptive modulation is traditionally based on the CSI feedback from the receiver, which is not available for LR-UWAC. Hence, in contrast to traditional adaptive modulation, we propose a *blind adaptive modulation* technique.

Rather than setting the modulation scheme directly based on the performance calculation of the predicted channel, we take a channel classification approach and propose three methodologies:

- 1) *Throughput prediction*: predicts the BER of each modulation scheme according to the predicted channel, and then compares their efficient throughput;
- 2) *Feature classification*: classifies the predicted channel based on the channel features;
- 3) *Throughput classification*: classifies the predicted channel based on its performance on the different modulation schemes.

Clearly, if the environmental information is accurate, the direct approach would yield better performance compared to our suggested approaches in terms of matching the right modulation to the channel. However, our evidence shows that when this information is only slightly biased or noisy, the performance of the direct approach degrades, while the performance of our approaches remains stable. This robust behavior is due to the abstract layer of classification, which is able to lock onto the channel's important features [17]. Hence, the contribution of our work is twofold:

- A blind adaptive modulation pre-setting method for LR-UWAC at the transmitter end. Instead of the traditional approach for adaptive modulation that builds on feedback from the receiver, our approach adapts the modulation by a prior evaluation of the channel. By training a classifier using an acoustic propagation model, we lock onto the channel's important features, thereby reducing the sensitivity to mismatches in the environmental information.
- A methodology to classify LR-UWAC channels into a fixed set of channel types. We classify different channel types by their performance of different modulation schemes.

We run intensive numerical simulations to compare the performance of the three approaches with the direct approach. The results show that the throughput classification approach achieves the best performance, with an average accuracy of

around 80% in matching the best modulation scheme to the channel. To demonstrate the proposed approaches in a real environment, we performed a long-range sea experiment, where transmission was made from a single source simultaneously to receivers located at ranges of 100 km and 50 km from the transmitter, and different depths for the transmitter and receivers were tested. This setup resulted in 15 different channel impulse responses; for all, the results show that the throughput classification approach provides the most accurate prediction of the best modulation scheme.

The remainder of this paper is organized as follows: A survey of previous works on LR-UWAC and adaptive modulation is provided in section II. In section III, channel modeling, support vector machine and the effect of the underwater acoustic channel on communications are discussed. In section IV, our three channel classification approaches and the direct calculation benchmark for adaptive modulation are introduced in detail. In section V, simulation and sea trial results are discussed. Concluding remarks are drawn in section VI.

II. STATE OF THE ART

A. LR-UWAC Systems

Early in 1993, Stojanovic *et al.* [18] demonstrated adaptive multi-channel combing and equalization for a distance of 203 km. The source level was 193 dB re 1 μ Pa/V @ 1m and the receiver consisted of a 12-element vertical array. The results showed error-free communications for a data rate of 660 bps. The experiment was conducted in deep water, and the receiver was placed in the acoustic convergence zone. The longest distance LR-UWAC was reported in 2001 by Freitag and Stojanovic [11], who analyzed the ATOC tomography signal in order to estimate the rate and reliability of LR-UWAC. By combining signals from 20 hydrophones, communication with a data rate of 37.5 bps was achieved for a distance of 3250 km across the North Pacific Ocean from Southern California to the Hawaiian Islands.

In recent years, more LR-UWAC sea trials have been conducted, and different modulation schemes were tested. Non-coherent modulation, like spread spectrum (SS) [19]; higher order of phase modulations, like QPSK [12]; and multi-carrier modulation, like OFDM [13] were tested for distances between 550 km and 700 km in deep water off the Southern California Coast. In addition, LR-UWAC with 16-QAM was achieved for a distance of 600 km in west of Izu-Ogasawara Islands [14]. A 102 m aperture vertical array comprised of 18 elements was used in the sea trial, and a data rate of 400 bps was achieved.

These typical works on LR-UWAC in recent years are listed in Table I. In order to compensate for the high transmission loss due to long-range propagation, arrays of multi-hydrophones were used. The reported sea trials were all conducted in deep water, while LR-UWAC in shallow water is seldom reported.

B. Adaptive Modulations

In this work, we consider our work as an *adaptive modulation* scheme, and focus only on the adaptation of the

TABLE I
A SUMMARY OF RECENT WORKS ON LR-UWAC

researcher	carrier frequency	modulation	data rate	sea depth	range	sound level	receiver
Stojanovic [18]	1000 Hz	QPSK	660 bps	2000 m	203 km	193 dB	VLA (12 elements)
Freitag [11]	75 Hz	BPSK	37.5 bps	~4500 m	3250 km	195 dB	VLA (20 elements)
Song [12]	250 Hz	QPSK	50 bps	4500 m	550 km	172 dB	HLA (64 elements)
Kang [13]	250 Hz	OFDM	7.5 bps	4500 m	700 km	172 dB	HLA (64 elements)
Liu [19]	200 Hz	SS	6.4 bps	4500 m	550 km	172 dB	HLA (64 elements)
Shimura [14]	500 Hz	16-QAM	400 bps	4000 m	600 km	200 dB	VLA (18 elements)

modulation scheme at the transmitter only. Adaptive modulation at the transmitter was first introduced in radio frequency (RF) terrestrial communications for choosing between QAM and OFDM to enhance spectral efficiency while maintaining a target error performance over fading channel by changing the constellation size of QAM [20] or the subcarrier selection of OFDM [21]. Adaptive modulation is mostly explored by pre-computing the SNR region boundaries [22], [23]. Here, feedback from the receiver regarding the channel state information (CSI) is needed. In [24], an automatic modulation classification algorithm is proposed to lower the amount of signaling overhead and to gain flexibility. Based upon the maximum-a-posteriori (MAP) criterion, a classification method for digital QAM schemes is developed to exploit the channel reciprocity and the receiver knowledge about the overall transmission data rate.

As an effective method to enhance spectral efficiency, adaptive modulation is also introduced to UWAC. For example, in [25], a single-carrier phase shift keying (PSK) based adaptive modulation system was proposed, where both the constellation size and the turbo code rate were adjustable according to an index of achievable information rate with independent, identically distributed (i.i.d.) Gaussian inputs and post-equalization SNR. In [15], OFDM adaptive modulation was used to maximize the system throughput under a target average BER. Both the modulation level and the power of the OFDM subcarriers were adjusted, and the method was demonstrated in a sea experiment. Another option is to change the modulation scheme completely. In [16], the effective SNR computed after channel estimation and channel decoding was defined as the performance metric to switch among a fixed number of transmission schemes.

Recently, machine learning was adopted to control the adaptive modulation. In [26], the effective SNR was defined as the channel state; reinforcement learning (Dyna-Q) was used to predict the channel state and calculate communication throughput based on the real and simulated transmission experiences of data communications. Then, based on the channel state and the calculated throughput, the algorithm chooses the modulation order. A similar idea was suggested in [27], where channel parameters were evaluated by tracking the channel's dynamics. This allows convergence to a modulation that minimizes the system cost without perfect knowledge of the CSI.

In [15] and [28], real-time adaptive modulation experiments were presented. The receivers were buoys with RF antenna for transmitting back the CSI. To facilitate adaptive modulation when RF is not available, acoustic transmission is used for

CSI feedback instead. While coordination with the receiver is effective in obtaining the CSI, due to the channel's time variation and the slow propagation in the water, in the context of UWAC such CSI may arrive to the transmitter as outdated information. Furthermore, this feedback increases the system's overhead - both in terms of delay and energy. Naturally, these problems compound in LR-UWAC, where the receiver may not even have the means to transmit a feedback message. Thus, while inspired by the concept of adaptive modulation, we do not require feedback from the receiver. Instead, we update the modulation based on knowledge available only at the transmitter, and tolerating the possibly mismatched information by adding an abstract layer of channel classification. Therefore, our method highly depends on the accuracy of the underwater acoustic propagation model.

III. PRELIMINARIES

A. Underwater Acoustic Channel Modeling

Recognized as one of the most complex communication mediums, underwater acoustic channel is characterized by a strong multi-path propagation, significant Doppler shifts and a high power attenuation. For UWAC, modeling the underwater acoustic channels provides an effective approach for channel analysis by simulating the influence of real water environment parameters on underwater acoustic channels. In general, the acoustic propagation model is aimed to efficiently solve the wave equation by restricting solutions to the frequency domain or by numeric simplifications. The propagation models can be categorized into five techniques: ray theory, normal mode, wavenumber integration, fast field and parabolic equation [29].

Several open-source models are available for modeling underwater acoustic propagation [30], each of which of unique applicability domain. The analyst is normally faced with a decision matrix involving water depth (deep versus shallow), frequency (high versus low), and range-dependence (range-independent versus range-dependent ocean environments). As we focus on LR-UWAC for deep and shallow water, we find that both ray theory and PE are most suitable. Comparing the two models, we argue that the result of ray theory is less applicable for our needs than the PE model. This is because, ray tracing is less accurate near the sea boundaries where the assumption of a plane acoustic wave breaks. Thus, since in both our simulations and experiments we explore performance when the transmitter and receiver are also near the sea surface and bottom, in the following we use the PE model. Yet, our approach can fit also other propagation models.

The PE model approximate the solution for the wave equation via factoring the outgoing wave components, and generalizing very-wide-angle parabolic equation [31]. It is very effective for solving range-dependent ocean acoustics problems.

PE is derived from the standard 2-D Helmholtz equation in cylindrical coordinates:

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r} \frac{\partial p}{\partial r} + \frac{\partial^2 p}{\partial z^2} + k_0^2 n^2 p = 0, \quad (1)$$

where $p(r, z)$ is the acoustic pressure, $k_0 = \omega/c_0$ is a reference wavenumber, and $n(r, z) = c_0/c(r, z)$ is the index of refraction. Factoring the operation in Eq. (1),

$$\left(\frac{\partial}{\partial r} + ik_0(1+X)^{1/2}\right)\left(\frac{\partial}{\partial r} - ik_0(1+X)^{1/2}\right)p = 0, \quad (2)$$

$$X = k_0^{-2}\left(\rho \frac{\partial}{\partial z} \frac{1}{\rho} \frac{\partial}{\partial z} + k^2 - k_0^2\right), \quad (3)$$

Neglecting the back-scattered energy, the outgoing wave equation can be obtained

$$\frac{\partial}{\partial r} = ik_0(1+X)^{1/2}p. \quad (4)$$

Applying an n -term rational function to approximate the exponential function, the solution of Eq. (4) is

$$p(r + \Delta r, z) = e^{ik_0 \Delta r} \prod_{j=1}^n \frac{1 + \alpha_{j,n} X}{1 + \beta_{j,n} X} p(r, z). \quad (5)$$

Based on the split-step Padé solution, it allows large range steps and is very efficient. Range dependence is handled accurately by applying an energy-conservation correction as the acoustic parameters vary with range. An initial condition (or starting field) is constructed using the self-starter, which is an accurate and efficient approach based on the PE method.

B. The Effect of the Underwater Acoustic Channel on Communications

Long-range underwater acoustic channels are characterized by a non-linear power attenuation and a long delay spread of sometimes seconds [14]. Unlike short-range underwater acoustic communication, for which an assumed system model like the tap-delay line can be used for channel equalization, in LR-UWAC, the propagation is modeled by acoustic beams. In terms of communications, this leads to unresolvable channel taps. In addition, since the long-range underwater acoustic channel is highly sensitive to environmental parameters - such as the bathymetry, the sound speed profile (SSP), the surface status and the bottom characteristics - channel equalization should be efficient. Moreover, the propagation characteristics highly depend on the locations and depth of the transmitter and receiver.

To demonstrate the dependency of LR-UWAC channels, in Fig. 1 we show two channels measured during a long-range sea experiment in the South China Sea. The channels were measured simultaneously at depths of 40 m (channel 1) and 60 m (channel 2). The sea depth was 90 m while the transmitter was deployed at the depth of 30 m, and the transmission distance was 80 km. The measured SSP for

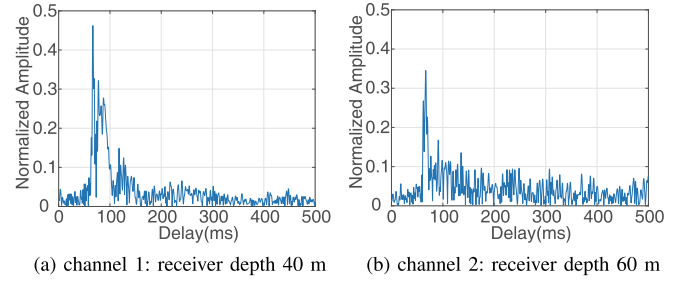


Fig. 1. An example of measured channel impulse responses from a sea experiment for transmission over 80 km range.

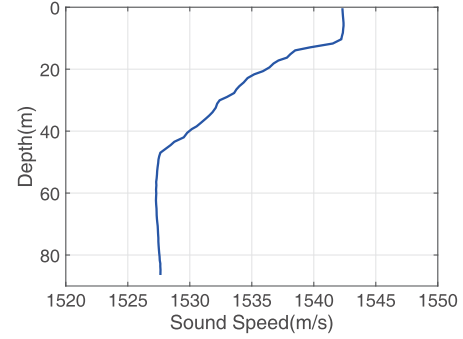


Fig. 2. SSP measured in South China Sea experiment.

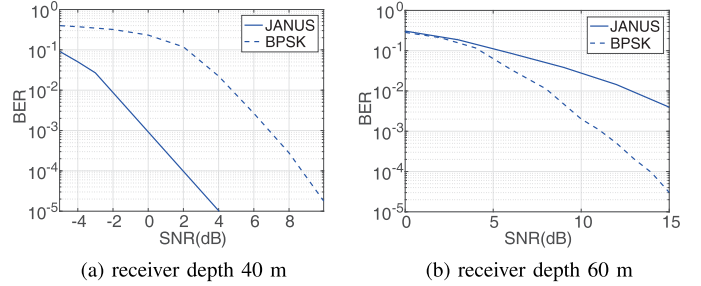


Fig. 3. An example of BER performance vs. SNR under different channels.

this experiment is shown in Fig. 2. Even for such a small change in the configuration, we observe that the delay and multi-path structure are significantly different between the channels, possibly due to the temperature layer in roughly 50 m. This change may very well be since the receivers depth is directly at the two sides of the thermocline layer shown on the SSP. Between a choice of JANUS [32], for incoherent frequency-hopping stable modulation, and BPSK, for coherent faster rate modulation, observing the two channels, our goal is to choose the best modulation for the two channels. To quantify this choice, in Fig. 3 we compare the BPSK performance with JANUS for the two channels. JANUS was used over a 940 Hz - 1460 Hz band, divided into 26 sub-bands for frequency-hopping modulation, and the data rate is 20 bps. BPSK was used over a frequency range of 900 Hz - 1500 Hz, and a data rate of 234 bps. Noise is added to calculate the BER for different SNR levels. The results show that BPSK outperforms JANUS for channel 2, and vice versa for channel 1. This is because the multi-path length of channel 2 is much longer than channel 1. It spans over more than 7 JANUS symbols, which albeit JANUS frequency hopping

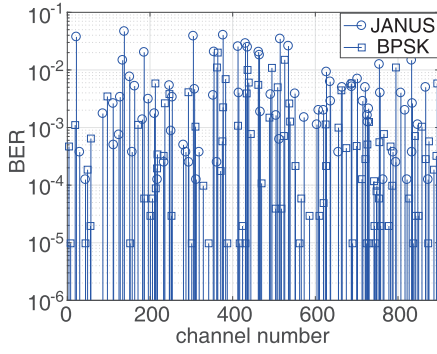


Fig. 4. BER performance of JANUS and BPSK for the 900 different simulated channels. The best modulation type changes across the channels.

characteristics, introduces strong ISI. The BPSK receiver can reduce such interference using a long forward filter coefficient, which is something that JANUS does not include. Hence, the performance of JANUS degrades compared to BPSK. Thus, predicting the channel impulse response may greatly increase the communication performance.

A comprehensive analysis for the dependency of LR-UWAC performance is shown in Fig. 4. The figure shows the BER performance of 900 different simulated channels for JANUS and BPSK. The channels are simulated using the PE model. To generate a variety of simulated channels, we randomly selected various environmental parameters (SSP, sea depth and bottom parameters) and transmitter and receiver deployments, more details about the parameter setting will be described latter in Table II. To focus on the effect of the channels, no noise is added to the received signals. While the symbol energy of JANUS is higher, we still can observe that for some channels, JANUS's performance is worse. For other channels, BPSK is worse. We claim that this BER performance diversity motivates the need to match the modulation scheme and the channel type.

C. Support Vector Machine

In this paper, a machine-learning based method is used adaptively choose the modulation type at the transmission end. More specifically, it is a support vector machine (SVM) [33] based method. SVM is a useful technique for data classification. A classification task usually involves separating data into training and testing sets. Each instance in the training set contains one "target value" (i.e. the class labels) and several "attributes" (i.e. the features or observed variables). The goal of SVM is to produce a model (based on the training data) which predicts the target values of the test data given only the test data attributes.

Given a training set of instance-label pairs $(x_i, y_i), i = 1, \dots, l$ where $x_i \in R^n$ and $y \in 1, -1^l$, the support vector machines require the solution of the following optimization problem:

$$\begin{aligned} \min \quad & \frac{1}{2} w^T w + C \sum_{i=1}^l \xi_i, \\ \text{subject to } & y_i(w^T \phi(x_i) + b) \geq 1 - \xi_i, \\ & \xi_i \geq 0. \end{aligned} \quad (6)$$

Here training vectors x_i are mapped into a higher (maybe infinite) dimensional space by the function ϕ . SVM finds a linear separating hyper-plane with the maximal margin in this higher dimensional space. $C > 0$ is the penalty parameter of the error term. Furthermore, $K(x_i, x_j) \equiv \phi(x_i)^T \phi(x_j)$ is called the kernel function. The following are four basic kernels of SVM:

- linear: $K(x_i, x_j) = x_i^T x_j$,
- polynomial: $K(x_i, x_j) = (\gamma x_i^T x_j + r)^d, \gamma > 0$,
- radial basis function (RBF): $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2), \gamma > 0$,
- sigmoid: $K(x_i, x_j) = \tanh(\gamma x_i^T x_j + r)$.

IV. ADAPTIVE MODULATION APPROACHES

The stage is now set to present our adaptive modulation approach for LR-UWAC. The focus of our work is to adaptively choose between a given number of modulation approaches at the transmitter side. In the absence of accurate environmental information, it is hard to precisely predict the channel. Hence, rather than choosing the transmission rate or the parameters of the channel coding, which albeit affecting performance are sensitive to the exact form of the channel, we focus on the lowest level of setting only the modulation scheme. As a test case, we consider a set of modulation schemes, namely JANUS, BPSK, QPSK and OFDM, from which we choose the best one for a given environmental information. We choose these schemes because they represent incoherent modulation, coherent modulation with lower and higher data rate, and multi-carrier modulation, respectively. This said, our approaches can also be used for other modulation schemes. As a criterion to choose from among the modulation schemes, we pack the transmission data into frames, and calculate the efficient throughput of the communication system, defined by

$$S_e = S_0 \phi_p = S_0 (1 - \phi_b)^N, \quad (7)$$

where S_0 is the data rate of different modulations, ϕ_p is the probability of a frame being received correctly, ϕ_b is the BER, and N is the number of bits within a frame. We target the metric S_e , since it reflects both the transmission rate and the rate of communication success. We focus on the practical case that the CSI is either limited or mismatched, such that the direct decision from the maximum S_e of the four modulation schemes may lead to a wrong choice. To achieve a less sensitive choosing mechanism, we turn to channel classification and offer three approaches.

A. Throughput Simulation: Our Benchmark

Before we describe our methodology, we present our benchmark, which we refer to as the throughput simulation approach. The process of throughput simulation is illustrated in Fig. 5. The process starts with a set of measured or evaluated environmental parameters. These include the bathymetry map, typical SSP, roughness of the sea boundaries, depth and range of the transmitter and receiver, and carrier frequency used. These parameters are fed into the PE model to predict the sound pressure for a specific frequency in the given locations of the transmitter and receiver. By changing the input

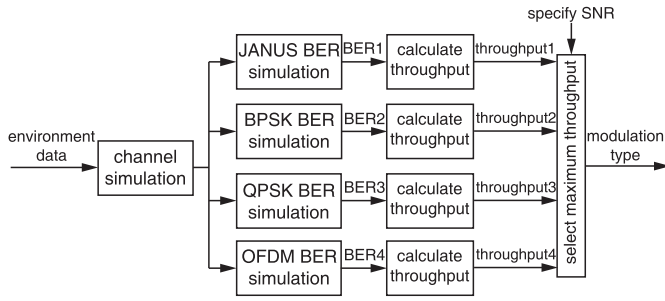


Fig. 5. Illustration of adaptive modulation by throughput simulation approach.

frequency for the prediction, we obtain the sound pressures within the whole communication bandwidth. These responses are then accumulated to form the channel's response in frequency domain. Converting this response to the time domain and convolving it with the transmitted signal, we obtain the received signal to each considered modulation scheme.

Once a received signal is obtained, the efficient throughput in (7) is calculated for the four modulation schemes. To factor in the effect of noise, Monte-Carlo simulations are performed, where the same information-bearing bits are modulated by different modulation schemes, the resulting signals are convolved with the predicted channel, and an additive noise is added. Finally, the calculated average efficient throughput of different modulation schemes are compared, and the one leading to the highest S_e is chosen for the transmission. Naturally, the accuracy of the throughput simulation scheme is directly affected by the accuracy of the environmental parameters and by the correctness of the propagation model. Hence, we refer to it as the *direct* approach.

To relax the sensitivity to the modeled channel in choosing the modulation scheme, we would like to make our decision according to the channel's most important features. Intuitively, this may serve as a filter for the decision. We thus turn to machine-learning classification to add an abstract layer between the predicted channel and the process of hard decision. Our three approaches for this are presented below.

B. Throughput Prediction

Our first channel classification approach tries to predict the BER of each modulation scheme directly by the channel impulse response. That is, in contrast to the direct approach, no demodulation is made. Fig. 6 illustrates the throughput prediction approach. BER prediction is obtained through support vector regression (SVR) [33]. The training step involves calculation of the throughput for each of the considered modulation schemes based on a large ensemble of simulated channels, while the prediction step performs the same operation on the predicted channels.

1) *Training*: With the absence of a large variety of channels from sea experiments, we train the SVR using simulated channels. These channel impulse responses are obtained via a PE model with random, but reasonable, environmental parameters. The channel impulse responses information we use are the time delays and amplitudes of each path. To achieve

a wide range of channels, we change these parameters, e.g., the transmission range, bathymetry map, sound speed profile, transmitter and receiver depths, and sea bottom characteristics. An example of different channel impulse responses obtained from such simulations is shown in Fig. 7. In the simulation, the sea depth is 140 m, and the transmitter and receiver are deployed to a 100 m depth in both cases. The only difference is the SSP, which is shown on the top-right of each sub-figure. Also the deployment is marked on the SSP figure, on the left side is the transmission depth, and the right side is the receiver depth. While the channels were made for the same transmission distance, we observe a significant change in their structure by only changing the SSP. Still, the basic structure of the two channels remains similar. The BER for each simulated channel and for each of the modulation schemes is predicted through a Monte-Carlo process. These BER values are used to train an SVR model.

2) *Prediction*: In the next step, upon being given an actual sea environment and the transmitter and receiver locations for LR-UWAC, the channel impulse response is calculated and its time-domain complex propagation pattern is used as a test input for the SVR model to produce the BER prediction for each of the tested modulations. Finally, the efficient throughput is calculated based on the predicted BER and the corresponding data rate, and is compared among the different modulation schemes to yield the preferred one.

C. Feature Classification

1) *Channel Type Classification*: Our second channel classification approach determines the class of the predicted channel by channel features. The process of the feature classification approach is illustrated in Fig. 8. The approach relies on the set of simulated channels as described in the throughput prediction approach. However, rather than estimating the BER, classification is made over a fixed set of channel types. Specifically, the various channels are hand-labelled to fixed types based on their features. Each channel type represents a different channel structure, which in turn, fits a different modulation scheme. For example, the channel in Fig.7a fits a QPSK modulation better because the multi-path pattern is simple enough for QPSK to function well at a higher bit rate. However, the channel in Fig.7b fits a BPSK modulation better as here the multi-path delay spread is longer. The set of simulated channels and the corresponding labeling are used as inputs to train a SVM that classifies time domain channel impulse response to channel types. Similarly, in testing mode, i.e., upon given a predicted time domain channel impulse response based on an actual sea environment and transmitter and receiver locations, the SVM predicts the channel type. The appropriate modulation is then determined by a look-up table that maps channel types to modulation schemes.

2) *Setting the Number of Channel Types*: To set the number of channel types, we assess the differences between the simulated channels. Both the channel structure and the delay spread are important features for channel type classification. However, since the former is hard to quantify, to determine the number of types in a quantitative fashion, we explore

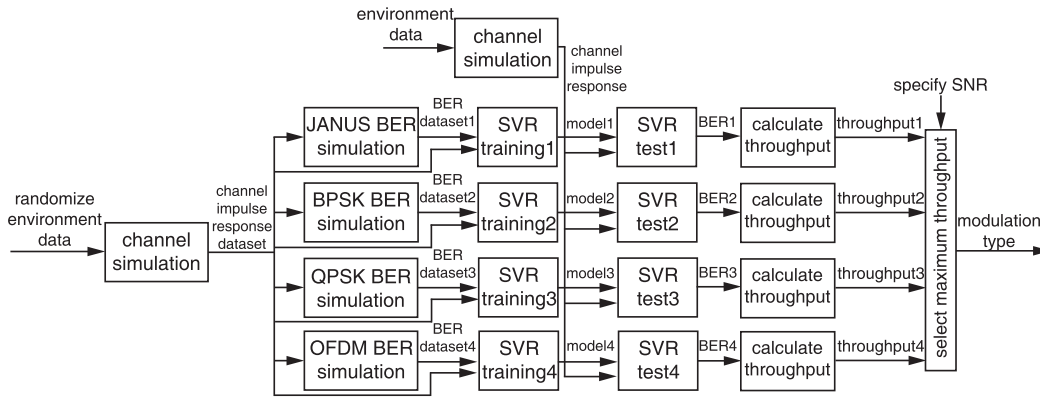


Fig. 6. Illustration of adaptive modulation by throughput prediction approach.

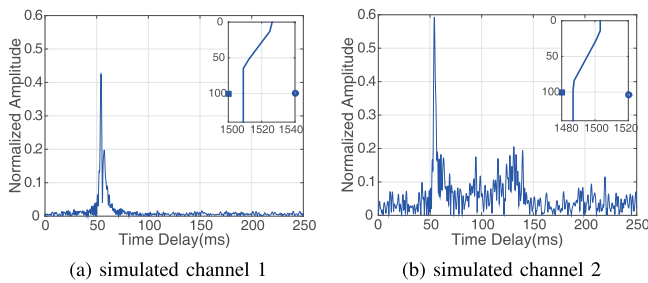


Fig. 7. An example of simulated channels generated by the PE model with similar environmental parameters. A significant change is observed between the channels.

the best division of the available channel's dataset through the channel's delay spread. We test a certain set of types using the K-means [34] classification, and calculate a principle component analysis (PCA) [35] for the set of delay spread measurement for the classification output from the K-means. This yields a sorted vector of eigenvalues of the covariance matrix. PCA projects the measurement into a set of unrelated components, which can be perceived as different channel types. The eigenvalues represent the contribution of different types on the whole measurement. Thus, we determine the number of channel types based on the number of dominant eigenvalues.

The result of the PCA analysis for the same set of 900 simulated channels, also explored in Fig. 4, is shown in Fig. 9. We observe that when the number of types exceeds four, the minimum component value is less than 10% of the maximum component value. This means that the choice of four channel types covers most of the channel features. This choice is expected to balance between channel diversity and the number of channels within each channel type.

3) *Channel Labeling*: We label the channels according to their features, especially the delay spread and the multi-path structure, which affect the demodulation process. In Fig. 10, we show examples of the four types of simulated channels. We make the following observations:

- 1) Type 1 has a simple multi-path structure with only a few strong paths and a short delay spread.

- 2) Type 2 has a strong arrival in terms of amplitude located at the beginning of the channel impulse response.
- 3) Type 3 is of an inverse structure to Type 2. That is, less significant multipath which shows after the main one (in terms of amplitude) in Type 2 are now shown before the main path. This type of channel can be obtained when a set of multipath arrivals are received within the main lobe of the direct path, leading to negative superposition that suppresses the direct path compared to a later arrival.
- 4) Type 4 shows a complicated structure with a long delay spread and with no significant multi-path components.

After labeling the channels, we map the four types to the four modulation schemes we considered in this paper. As OFDM is a multi-path-tolerant modulation scheme, we assign it to channel Type 4. Type 1 has the most simple multi-path structure, so we assign it to QPSK, which has the highest data rate. Compared to Type 2, a BPSK receiver of channel Type 3 will be designed with a longer feedback coefficient and a shorter forward coefficient, which will take long time for the demodulation. But for JANUS, there is not much difference between the demodulation time for both channels. Hence, we assign BPSK to Type 2, and JANUS to Type 3.

As a measure of channel classification, we choose SVM because it is able to capture dominant features of the classified data using a relatively small number of input instances [33]. We explored four optional kernels for SVM: linear, polynomial, RBF, and sigmoid. Classification results in terms of the accuracy in labeling channels for all four kernel types is shown in Fig. 11. We observe that RBF produces the best accuracy (75.99%). However, an improved version of the RBF kernel, referred to as *RBF CV*, performs even better. Here, the training channels are separated into five subsets, each subset includes one fifth of the training channels, and cross validation is used to obtain the best accuracy (83.17%). The RBF CV kernel is thus used in the following simulations.

D. Throughput Classification

Our third channel classification approach classifies the channel based on the performance of the different modulations, to directly map between channel types and modulations. The

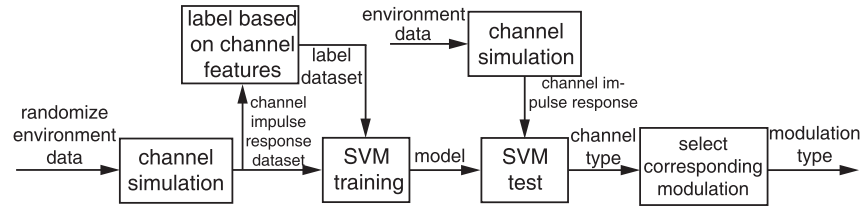


Fig. 8. Illustration of adaptive modulation by feature classification approach.

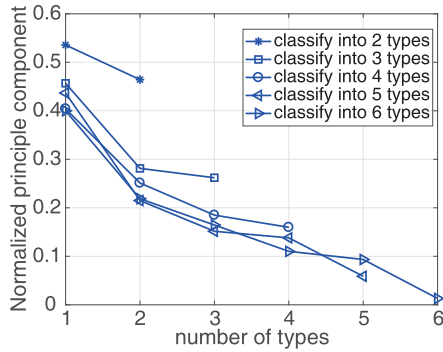


Fig. 9. PCA results for classifying the channels into a different number of types. The principle component of each curve is normalized. With a smaller value of principle component, it is less significant to form a new type.

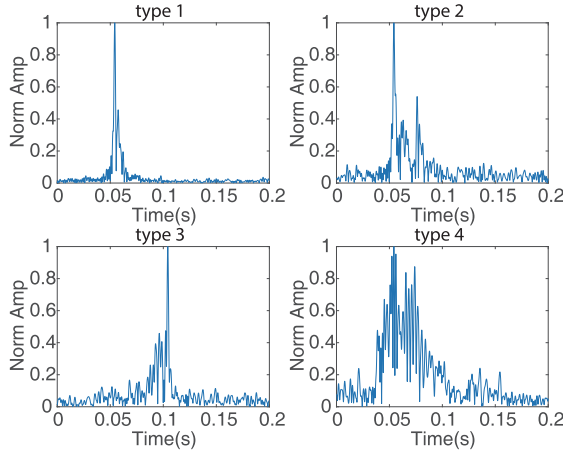


Fig. 10. Typical structures of the four assigned different channel types.

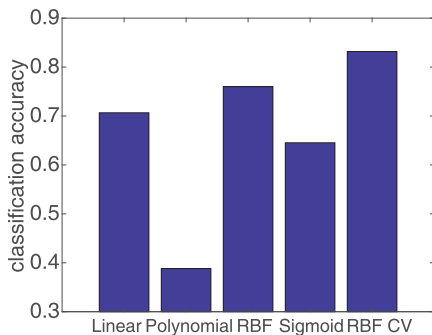


Fig. 11. Classification accuracy of different kernel types.

process of this approach is illustrated in Fig. 12. The input for the SVM training is the set of channel impulse responses and the corresponding best modulation type. The best modulation

for each channel is determined by the one that leads to the highest throughput. The trained model is used in the testing mode to project a given channel impulse response to the best modulation to use.

E. Discussion

The underwater acoustic domain is a complex environment, and the communications performance may be effected by other factors but the linear channel impulse response as considered here. For example, Doppler shift can distort the received signal yielding to a preference of a Doppler-tolerant modulation scheme. While we prefer to keep our assumptions about the state of the receiver to minimum, we note that our scheme is flexible enough to encompass other information than the channel if available. For example, if the velocity of the receiver relative to the transmitter is known, an interpolation during the e.g., BER calculation can reflect the effect of the Doppler shift.

Among the above four evaluation approaches, we note that the throughput simulation approach is the simplest to use. However, due to the complexity of the long-range acoustic channel, and the expected mismatch between the available environmental information at the transmitter and the actual channel condition, we expect the channel prediction error may lead to a wrong decision in terms of the best modulation scheme to use. Alternatively, the three above channel classification approaches use machine-learning to indirectly find the relation between the channels and the metrics for the decision-making among the available modulation techniques. This indirect approach allows a better tolerance for mismatches in the channel prediction, thereby leading to robustness to environmental conditions. In the following section, we analyze the performance of the four adaptive modulation approaches in both numerical simulation and sea experiments.

V. PERFORMANCE ANALYSIS

A. Numerical Simulations

In this section, we evaluate the performance of our three channel classification approaches and the direct calculation benchmark for the adaptive modulation for LR-UWAC via numerical simulations. The simulations involve a simulated channel's dataset, generated by the PE model. The simulated channels are created by randomly choosing the environmental parameters. As we aim to use a specific transmitter to achieve LR-UWAC in shallow water, the frequency range is fixed at 900 Hz - 1500 Hz, and the range is fixed at 100 km. However, the approaches can also be used for other frequency ranges and

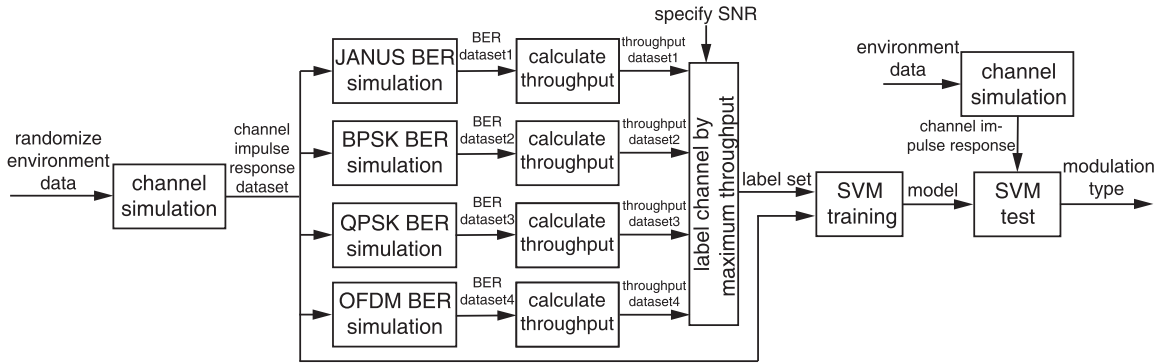


Fig. 12. Illustration of adaptive modulation by throughput classification approach.

TABLE II
CHANNEL SIMULATION PARAMETER SETTINGS

Parameter	Setting
frequency	900 Hz-1500 Hz
range	100 km
bathymetry	range-dependent sea depth with an average depth randomly chosen from 70 m to 170 m
sound speed profile	zero or negative gradient SSP is considered. sound speed on the surface is chosen from a uniform distribution between 1500 m/s and 1540 m/s, the sound speed gradient is chosen from a uniform distribution between 0 s^{-1} and -0.3 s^{-1} .
depths of tx and rx	random distribution in the water column but with similar depth
bottom parameters	randomly chosen from the following four types 1: $c=1780 \text{ m/s}$ $\rho=2.10 \text{ g/cm}^3$ $\alpha=0.33 \text{ dB}/\lambda$ 2: $c=1680 \text{ m/s}$ $\rho=1.90 \text{ g/cm}^3$ $\alpha=0.46 \text{ dB}/\lambda$ 3: $c=1620 \text{ m/s}$ $\rho=1.80 \text{ g/cm}^3$ $\alpha=0.39 \text{ dB}/\lambda$ 4: $c=1580 \text{ m/s}$ $\rho=1.65 \text{ g/cm}^3$ $\alpha=0.28 \text{ dB}/\lambda$

distances. The average water depth varies from 70 m to 170 m. In each case, the water depth is range dependent; the difference of depth is less than 10 m. The sound speed on the surface is randomly chosen from 1500 m/s to 1540 m/s. SSP with a negative or zero gradient is more common in reality, so we set the SSP with a random gradient smaller than or equal to zero. In addition, we place both the transmitter and receiver with a random, but similar depth. The three sea bottom parameters, namely density (ρ), attenuation (α) and sound speed (c), are usually related. We select four sea bottom types, ranging from a hard bottom to a soft bottom. In the simulation, we randomly pick one of these four sea bottom types. The environmental parameters setting is listed in Table II. For our investigation, we generate 900 simulated channels from such a setting.

Recall that we consider JANUS, BPSK, QPSK and OFDM as the fixed modulation schemes to choose from at the transmitter end. The settings of these four modulation schemes and the hand-labeled assignment to channel types are listed in Table III. The efficient throughput calculation for the four modulations over the entire 900 simulated channels is presented in Fig. 13. To focus on the channels' effects, no noise is added to the received signals. We observe a high channel dependency for the results. Next, in Fig. 14, we show the results of the hand-labeled channel assignment of each of the

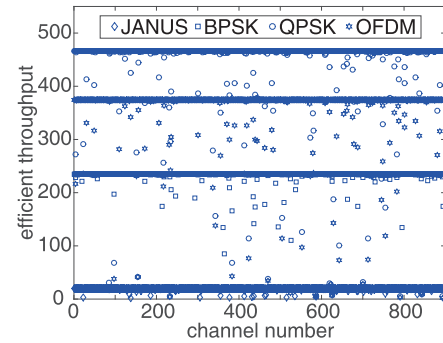


Fig. 13. Efficient throughput performance of the four modulations for the 900 different simulated channels without noise.

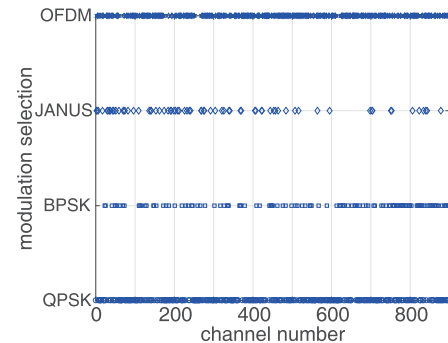


Fig. 14. Hand-labeling of the 900 simulated channels to the four modulation schemes.

900 simulated channels to the four modulation schemes. The calculated throughput and the labeled types are used to train our channel classifier.

We now compare the performance of our channel classification approaches with the direct calculation benchmark. Recall that the motivation for our approach is the absence of accurate environmental information at the transmitter and the lack of feedback from the receiver. This leads to mismatches between the model-based simulated channel and the true channel. To simulate this scenario, we simulated biased environmental information. That is, for a given set of environmental parameters and using the PE model, we create a *true channel*, and add bias to the environmental parameters

TABLE III
MODULATION SCHEME SETTINGS AND LABELING TO CHANNEL TYPES

scheme	JANUS	BPSK	QPSK	OFDM
frequency range	940 Hz - 1460 Hz	900 Hz - 1500 Hz	900 Hz - 1500 Hz	925 Hz - 1475 Hz
symbol duration	50 ms	3.33 ms	3.33 ms	1 s
data rate	20 bps	234 bps	466 bps	375 bps
demodulation method	official code [32]	DFE [36]	DFE [36]	LS [37]
channel type	Type 3	Type 2	Type 1	Type 4
remarks	—	1024 bits per frame 15% training	1024 bits per frame 15% training	1/2 pilot guard interval: 1/3 s

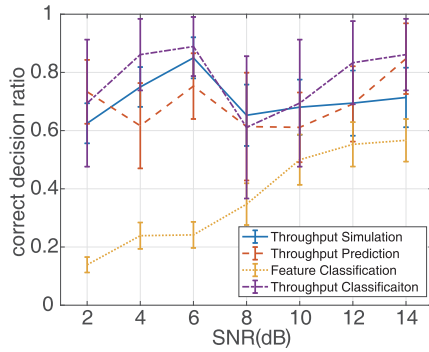


Fig. 15. Comparison of the correct decision ratio as a function of SNR for the four considered adaptive modulation approaches.

to create a *mismatched channel*. The former is considered the actual channel between the transmitter and the receiver, while the latter is the channel assumed at the transmitter end according to its mismatched environmental information. This mismatch is created by adding random bias to the SSP and to the bathymetry map. For the SSP, the bias is a random value uniformly distributed within $[-V_0/10, V_0/10]$, where V_0 is equal to $(V_{max} - V_{min})$, and V_{max} is the maximum sound speed, and V_{min} is the minimum sound speed in the water column. For bathymetry, we add a bias as a random value uniformly distributed within $[-D_0/10, D_0/10]$, where D_0 is equal to the maximum sea depth. We tested 40 true channel configurations; for each channel we added 10 randomly-chosen biases such that in total 440 channels were explored. Out of these channels, 40 were true channels and 400 were mismatched channels.

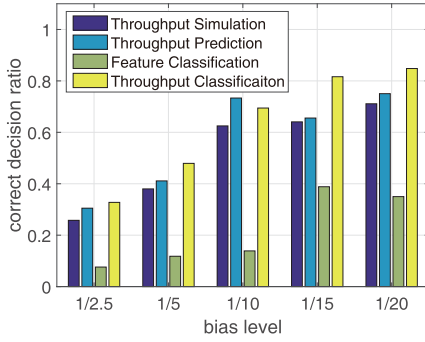
To measure the performance of the four approaches, we consider the correct decision ratio as a performance metric for each of the 40 true channels. This is defined as $N_{correct}/N_{total}$, where N_{total} is the total number of mismatched channels generated from one true channel (in our simulations, $N_{total} = 10$), $N_{correct}$ is the number of correct decisions within the N_{total} channels. This metric tests the accuracy level in picking the right modulation for the considered channel. The correct choice of the modulation scheme is determined by the calculated efficient throughput for the true channel, while the tested choice of the modulation is set by the mismatched channels. That is, for each of the 40 correct choices, we obtain 10 tested choices. The average correct decision index results and the standard deviation for the correct choices are shown as a function of the SNR in Fig. 15. Results

are shown for each of the four tested adaptive modulation approaches. As the feature classification approach classifies the channels by their multi-path features, it has the same result for different SNR, while the decision of the true channels varies with SNR. We can see the performance of feature classification depends on the SNR. For the remaining three approaches, SNR is considered one of the factors in the process, and the average accuracy is better. We also observe that the throughput classification approach achieves the best accuracy for all of the SNR, except 8 dB. It is higher than 70% in most of the cases; the average accuracy across all the SNR is 78%. Compared to the throughput simulation approach, the throughput classification approach is 5%-10% better in most cases. This shows the robustness of classifying channels into different types using a machine-learning method, rather than simply calculating the throughput on a single predicted channel when there is mismatch in the channel's prediction. Hence, the throughput classification approach can be a promising method for adaptive modulation schemes for LR-UWAC.

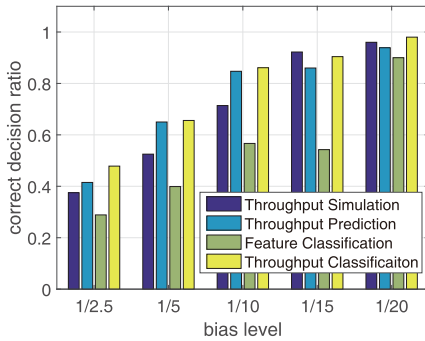
In Fig. 16, we show performance comparison in terms of the correct decision ratio for the four approaches as a function of different levels of biases added to the SSP. The biases are added in the same way as Fig. 15, but instead of setting the bias level to 1/10 of the original value, there are five different levels. Results are shown for SNR levels of 2 dB and 14 dB. As expected, for both explored SNR levels, a trend exists where the correct decision ratio decreases as the bias increases. Observing the results, we note that all four approaches can handle small bias of 1/20. In terms of performance decrease, best resilience for larger biases is shown for the Throughput Classification approach. This is because, rather than making decisions based on the simulated or predicted throughput, the abstract layer of classification in this approach tries to lock onto the channel's important features and is thus less sensitive to mismatches. Still, all these approaches performance decrease much when the bias is large (1/2.5). This result emphasizes the affect of mismatches in the SSP over the channel impulse response.

B. Experimental Results

1) *Experiment Setup*: Our three adaptive modulation approaches rely on environmental data to predict the channel impulse response. While we explore the effect of mismatched data, it is important to evaluate the performance in a realistic environment. To this end, we performed a designated



(a) SNR 2dB



(b) SNR 14dB

Fig. 16. Comparison of the correct decision ratio as a function of different levels of bias for the four considered adaptive modulation approaches.

LR-UWAC sea trial across the shores of Northern Israel. The experiment was conducted in March 2019; the location of the different nodes is shown in Fig. 17. The transmitter was deployed roughly 7 km west of Haifa, Israel, where the water depth is around 60 m. We used the Geospectrum M21-320 low-frequency projector with a matching M620-103 power amplifier, which is able to generate 201 dB re $1\mu\text{Pa/V}$ @ 1m power with a frequency range of 600 Hz to 1500 Hz. The depth of the projector changed during the experiment between 20 m, 35 m and 50 m. To explore different sea environments, we used two receivers: (1) one was located 100 km northwest from the transmitter, at a water depth of roughly 1800 m; a self-made acoustic recorder, based on the TASCAM DR100mk recorder and two Geospectrum M30-100 hydrophones, was deployed in 200 m of water. (2) The second receiver was deployed 50 km south of the transmitter, in a similar water depth as the transmitter (roughly 60 m). At this site, a similar recorder to that in the deep site was deployed in the middle of the water column. Recording was continuously carried out during the experiment such that no synchronization was needed among the three vessels.

Fig. 18 shows the assumed bathymetry profile from the transmitter to the two receivers. The ground from the transmitter to the shallow site was assumed to be a rough surface of sand, while that of the deep channel was assumed to be comprised of silt ground. The sea state scale during the trial was less than 2, so the sea surface roughness effect was

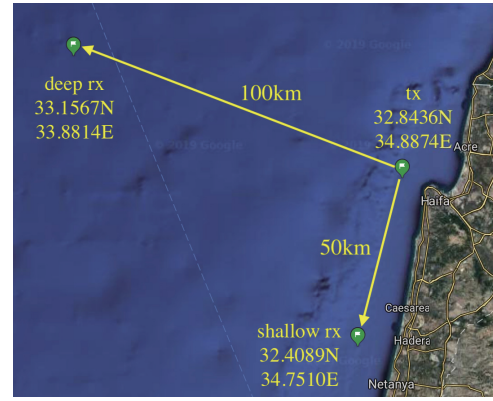
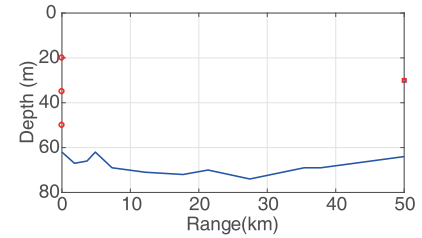
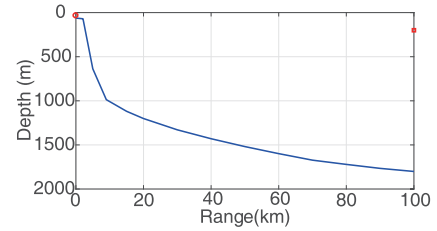


Fig. 17. Locations of transmitter and receivers in the experiment.



(a) transmitter to the shallow site



(b) transmitter to the deep site

Fig. 18. Bathymetry profile from the transmitter to the two receivers.

ignored for channel modeling. We measured the SSP during the experiment, both at the transmitter site and the shallow water receiver site. However, this was not performed at the deeper site due to the water depth. The SSP for both sites are shown in Fig. 19. We observe that the sound speed did not change much between the sites with a difference of less than 1m/s. For the deep site, we used the annual mean profiles of the East Mediterranean [38]. As a mean SSP was used, we did not assume the SSP varies with range.

During the experiment, we transmitted five signal types, namely chirp, JANUS, BPSK, QPSK and OFDM. The chirp signal was used to measure the channel, while the rest of the signals were communication signals. The setting of the different signals is the same as in our simulation, which is described in Table III. In each frame, we transmitted 288 bits for JANUS, 2060 bits for BPSK, 4608 bits for QPSK and 2048 bits for OFDM. In total, 43 frames were received during the experiment, including 28 frames from the shallow water site and 15 frames from the deep water site. Transmission was made with the projector's full power. However, due to the drifting of the two receivers, and due

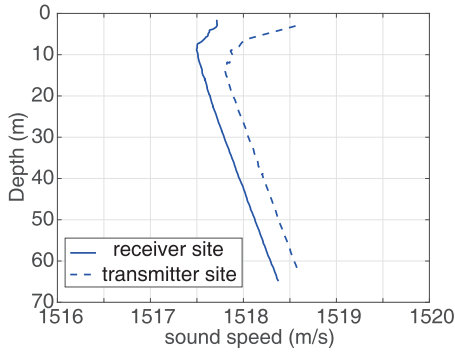


Fig. 19. SSP measured in the experiment.

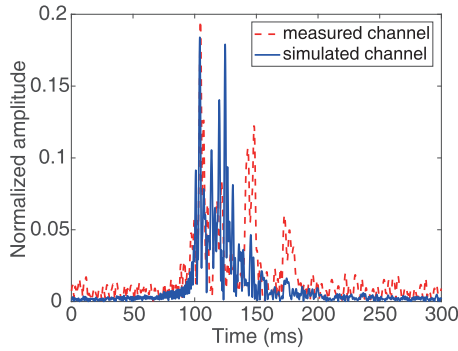


Fig. 20. A comparison between the predicted channel and the channel measured in the experiment.

to the time-varying channel, the signals were received at different SNR levels. The SNR range for different deployment is shown in Table IV. Performing a LR-UWAC experiment is a challenging, costly and time-consuming task. To make our work reproducible, we freely share the estimated channel impulse responses from the sea experiment at <https://drive.google.com/file/d/1g2sW7j3qvdTBnpj35CjS2ak83YQK0S80/view?usp=sharing>.

2) *Experiment Results:* We now discuss the results of the sea experiment. Fig. 20 shows an example of a measured channel impulse response recorded in the deep site for transmission at 20 m depth. In the figure, we also show the predicted channel impulse response obtained by channel modeling. While we observe some differences between the two channels, we argue that the channels have similar features. In particular, the two channels have long delay spread and significant main path, which makes both of them fit channel Type II. This rough match of the channels by pattern but not in terms of one-by-one channel types demonstrates that our approach is insensitive to environmental data mismatch.

In Table IV, we show the experiment's adaptive modulation results. For each SNR-depth combination, we calculated the efficient throughput of the four modulation schemes based on the measured BER, and chose the one with the maximum value as the most suitable choice. Then, we tested the four approaches and compared the results with the measured one. In Table IV, a $\checkmark$ means a correct match, whereas $\times$ refers to at least one frame of wrong modulation decision in the combination. We observe that throughput classification achieves the

TABLE IV
CORRECTNESS OF ADAPTIVE MODULATIONS IN DIFFERENT DEPTHS
AND SNR FOR THE FOUR APPROACHES

depth	SNR	throughput simulation	throughput prediction	feature classification	throughput classification
shallow tx 20 m	12 dB	$\checkmark$	$\times$	$\checkmark$	$\checkmark$
	14 dB	$\checkmark$	$\times$	$\checkmark$	$\checkmark$
shallow tx 35 m	2 dB	$\times$	$\checkmark$	$\times$	$\checkmark$
	4 dB	$\checkmark$	$\times$	$\times$	$\checkmark$
	6 dB	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
	8 dB	$\times$	$\checkmark$	$\times$	$\checkmark$
	12 dB	$\checkmark$	$\times$	$\checkmark$	$\checkmark$
shallow tx 50 m	8 dB	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
	12 dB	$\checkmark$	$\times$	$\checkmark$	$\checkmark$
	14 dB	$\checkmark$	$\times$	$\checkmark$	$\checkmark$
deep tx 20 m	2 dB	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
	6 dB	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
	8 dB	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
deep tx 50 m	4 dB	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
	8 dB	$\checkmark$	$\checkmark$	$\times$	$\checkmark$

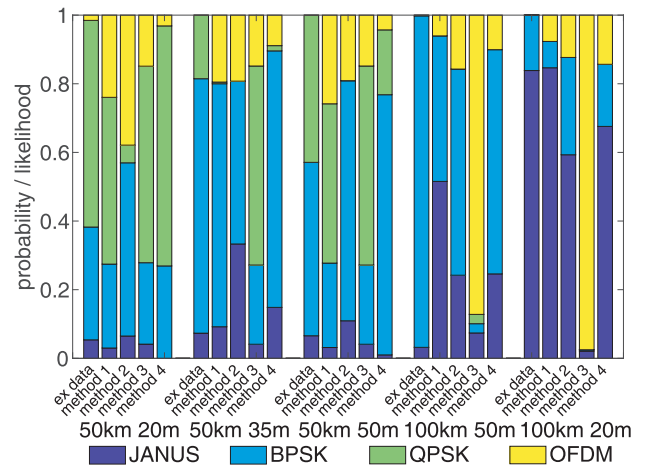


Fig. 21. Probability of the four approaches compared with experiment results at different depths and distances. Methods 1 to 4 refer to the four approaches introduced in this paper, namely throughput simulation, throughput prediction, feature classification and throughput classification.

highest accuracy, followed by throughput simulation, throughput prediction and feature classification. This result fits the conclusion from our simulation as shown in Fig.15.

Another interesting performance metric is *the likelihood of classification*: the dominance of choosing a certain approach over other approaches. A small difference may indicate a *chance* result, whereas a significant dominant choice will reflect the suitability of the chosen modulation scheme to the given channel impulse response. To measure this, we use the likelihood metric from the SVM predictions. Fig. 21 shows the normalized likelihoods in the choice of the preferred modulation scheme for the four adaptive modulation approaches. As a ground truth, the likelihood of the experiment data is shown in the first bar of each group. This is calculated by the normalized efficient throughput over the four modulation schemes. There are five combinations of depth and distance as shown in the five groups. To allow more diversity in the experiment data, in Fig. 21 the likelihood result for each of the channel groups was selected from a different SNR measurement. The selected SNR levels are 12 dB, 6 dB, 8 dB,

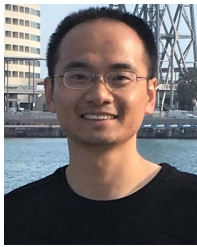
4 dB and 2 dB from the left to the right. This blend shows a different chosen modulation scheme for each of the five groups. In particular, QPSK is chosen for the first group, JANUS is chosen for the last group, and BPSK is chosen for rest of the groups. From the figure, we observe the advantage of the throughput classification approach, whose choices are made with a higher likelihood ratio testing.

VI. CONCLUSION

In this paper, we proposed three approaches of adaptive modulation schemes for LR-UWAC when feedback from the receiver is not available. For all approaches, a predicted channel impulse response, based on an acoustic propagation model, is used to make a decision aimed at maximizing the efficient throughput. Motivated by the concern that the environmental information held by the transmitter is possibly mismatched, we avoid the direct approach of making the decision according to a simulation over the predicted channel, and rather add an abstract layer that classifies the channel or predicts the channel's performance using machine-learning tools. We suggest it is a better solution by arguing that upon capturing the important features from the channel, a machine-learning-based classifier may have better resilience to mismatches in channel prediction. Our simulation and sea trial results show that our throughput classification approach provides improved performance in realistic channels, and leads to an accurate prediction result for the best modulation scheme, even for mismatched environmental information. In future research, we will extend this approach to more advanced machine-learning schemes.

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